Bruno Bispo dos Santos

Software Engineer & Computer Science Researcher · UNEB

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SUMMARY

Software Engineering student at UNEB with practical experience in systems engineering and full-cycle development: high-performance Rust applications, Tauri desktop software, scalable edge architectures on Cloudflare, and backend services with Spring Boot and Go. Passionate about compiler engineering, memory safety models and data precision — combining rigorous analytical discipline from corporate operations with modern software craftsmanship.

EXPERIENCE

Administrative Assistant

Nov 2020 — Aug 2022 · 1 yr 10 mos

Atacadão Pinto · Alagoinhas, Bahia, Brazil · On-site

Operational and commercial management responsible for database product catalog accuracy, market-driven dynamic pricing, inventory expiration risk mitigation, and promotional budget allocation.

- **Product Catalog & Database Integrity:** Managed precise data entry and ongoing updates of technical product details within the company database, ensuring 100% inventory accuracy and business continuity.
- **Pricing Optimization & Competitive Analysis:** Executed price adjustments and strategic pricing decisions based on market trends, competitive benchmarks, and internal fiscal policies, safeguarding competitiveness and profit margins.
- **Inventory Expiration & Cost Control:** Monitored stock turnover, price variations, and expiration schedules across thousands of SKUs to minimize financial write-offs and enforce regulatory compliance standards.
- **Promotional Budget Allocation & ROI:** Planned and allocated budgetary resources for discount campaigns and supplier promotions, optimizing cost efficiency and evaluating ROI to maximize volume and net margins.

Undergraduate Thesis — Generational References in Compilers

2026 — Present

Compiler Engineering Research · UNEB · Academic Research

Experimental evaluation of generational references as an automatic fallback when escape analysis cannot prove lifetime at compile time, benchmarked on the Arandu compiler.

- **Memory Model Design:** Formulated a decision tree bridging static ownership verification and runtime safety through generation-indexed slot handles in O(1).
- **Empirical Benchmarking:** Constructed comprehensive automated suites measuring acceptance rate, execution latency and heap overhead compared to standard Rust and C implementations.

Systems & Product Engineering

2026 — Present

Independent Software Engineering · Open Source & Remote

Architecture and delivery of production-grade systems: high-performance desktop applications in Tauri/Rust, scalable edge platforms on Cloudflare, and deterministic simulation engines.

- **Desktop & Tooling Delivery:** Built Imglyze (sub-millisecond parallel batch image processor) and HyCore Launcher (lightweight game client under 10 MB).
- **Cloud & Edge Infrastructure:** Designed Moussee.me with Cloudflare Workers edge routing, D1/R2 storage, custom domain decoupling and sub-100ms response targets.

SELECTED PROJECTS

Arandu

`Rust · Compilers · Cranelift · WebAssembly · Astro`

A modern ownership-based systems language.

I created the Arandu programming language and its native Rust compiler, built around static ownership verification, escape analysis and Cranelift code generation — with an in-browser WebAssembly playground and full documentation at arandu-lang.dev.

- Creator and author of the Arandu programming language and compiler in Rust
- Official platform with docs, specifications and in-browser WASM playground at arandu-lang.dev
- Static affine ownership semantics, escape analysis and memory safety guarantees
- Automated test suites and empirical performance benchmarks against C and Rust

Moussee.me

`SvelteKit · TypeScript · Cloudflare Workers · D1 · R2`

Identity and links for creators, at near-zero cost.

A high-performance link-in-bio platform with a Cloudflare-first architecture and full domain decoupling — a monorepo spanning a SvelteKit studio and an edge gateway.

- SvelteKit 5 admin panel with a visual WYSIWYG page editor
- Cloudflare Worker gateway for edge routing and custom domains
- Deterministic static renderer with critical CSS, plus a canonical theme catalog
- Zod data contracts, D1/R2 storage, CDN purge and analytics

Imglyze

`Rust · Tauri 2 · React · TypeScript`

Your images, lighter.

A local desktop app to optimize, convert and resize images in batch — with before/after comparison and processing that never leaves your machine.

- Parallel batch compression with per-format quality targets
- Import JPEG, PNG, WebP, BMP, AVIF — export the same set, plus rasterized SVG
- Reusable named presets and a virtualized queue for huge batches
- Light/dark themes, Portuguese and English, keyboard-first dialogs

Hycore

`Rust · Tauri · React · Tailwind CSS`

The launcher Hytale deserves.

An open-source, ultralight game launcher built on Tauri instead of Electron — native Rust performance, mod management, isolated profiles and sub-10 MB builds.

- Rust + Tauri core: tiny binaries, low RAM, smooth on modest machines
- Mod install/remove/update and multi-version management
- Isolated profiles ("Vanilla", "Magic Mods", "PVP")
- Discord Rich Presence and four languages (EN, PT-BR, ES, RU)

QuiJava

`Kotlin · Compose Desktop · Spring Boot · Hibernate · HSQLDB`

A desktop quiz platform.

An interactive platform to create and play quizzes, with authentication, ranking and history — a Material Design 3 desktop client over a Spring Boot backend.

- Jetpack Compose Desktop UI with Material Design 3
- Spring Boot + Hibernate/JPA persistence and Koin dependency injection
- Quiz creation, live gameplay with animated timer, ranking and history
- Structured as a real product: auth, profiles, scoring

EDUCATION

Bachelor of Software Engineering

2023 — Present

State University of Bahia (UNEB) · Campus II

Undergraduate degree focused on software architecture, systems, algorithms and data structures. Final thesis research in compiler engineering and memory safety models.

SKILLS

LANGUAGES

Rust · TypeScript · JavaScript · Kotlin · Java · Python · C · C# · Dart · R · SQL

FRONTEND & UI

React · Svelte · Astro · Tailwind · Flutter · Compose · HTML5 · CSS3

BACKEND & DATA

Spring Boot · Node.js · Express · PostgreSQL · Prisma · SQLite · Hibernate · gRPC · .NET

DESKTOP & SYSTEMS

Tauri · Unity · JavaFX · FXGL · Docker · Linux · Nix

TOOLS & INFRA

Git · GitHub · Cloudflare · Vite · Gradle · Neovim

COMPLEMENTARY COURSES

Administrative Routines Professional Certificate

2018

Concretta Escola da Construção

Professional qualification in business processes, commercial documentation, resource planning and administrative workflows.

Professional Computing & Systems Foundations

2013

Líder Cursos

Comprehensive foundation in computing systems, operational workflows and early programming curiosity that sparked my career path.

LANGUAGES

Portuguese `Native` English `Technical proficiency (fluent reading)`